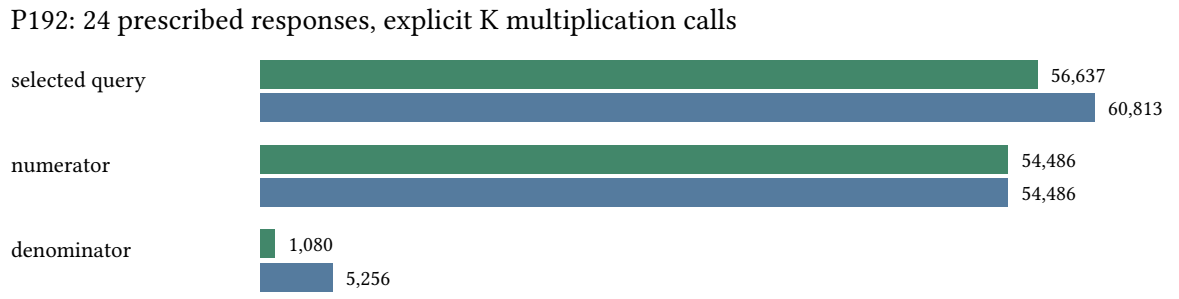
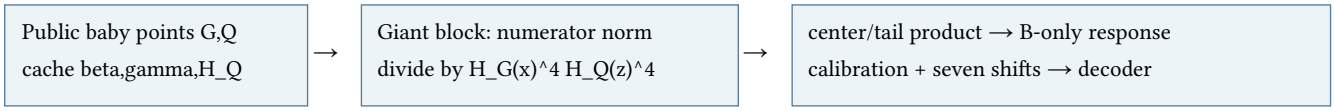
Cached denominator in the complete signed response



Green cached; blue legacy. Shared scale. Setup separate. Source: audit-summary.json and raw P192 rows.



P192 executes prescribed 27-factor windows; the complete decoder panel runs on the small curves.

Round66, 2026-11-09, T-1979. The selected signed baby/giant response now caches H_Q once per public target and evaluates its complete paired denominator by H_G(x(A))^4 H_Q(x(B))^4. The cached denominator feeds the rational numerator norm and the center/tail product, followed by the unchanged calibrated decoder. This qualifies the integrated component and its costs on the frozen inputs. Scientific standing is unverified producer evidence. [pcc:unverified]

Construction and proof. Round63 gives pair jet (E0+s*E1)/D^2 in K[Z]/H_G. Round64 supplies the singular-safe field norm of E0+s*E1. Round65 proves Norm(D^2)=H_G(x(A))^4 H_Q(x(B))^4. The denominator is nonzero by the nonrational-shift guards, so dividing both numerator coefficients preserves the marked first coefficient even when its constant coefficient vanishes. Each complete block is multiplied by its center jet and the blocks/tail are recombined. Changing the denominator implementation leaves the selected mathematical response unchanged. No numerator norm is inverted. [pcc:unverified]

Cache and custody. For each target Q, H_Q is constructed once from the existing public baby x-coordinates; it is reused across every giant block and query for that Q. The small-field implementation keeps b+1 additional field coefficients while processing one target. P192 keeps five extra coefficients at b=4, costing20 explicit K multiplication calls and0 inversions per target. Its qualifier retains all3 target caches, charging60 multiplications and15 cache coefficients in total; one-target construction needs one such cache. Its beta/gamma/H_Q storage is13 coefficients per target, source H_G another5. Reference graphs and temporary polynomials are additional qualifier memory, not hidden in these component slots. [pcc:unverified]

Frozen small controls. All381 calibrated cases and2694 coefficient queries pass. Both new denominator modes have byte-identical scientific outputs; every original B vector, B_cal, recovered point and decoder-count record matches sealed Round64. All11382 complete baby blocks are independently compared against direct paired factors; incomplete tails retain their original path. Selected query costs subtract complete-block direct validation work while including giant-point additions, numerator formation/norm, denominator formation/evaluation, both divisions and center/tail recombination. The qualifier still executes references, records their counters and includes them in its whole-process counters. [pcc:unverified]

Exact integrated small-panel counts, in (K multiplication including inverse powers, inverse, point-add calls), candidate versus legacy denominator: r7 (133036,5976,1400) versus (131836,5976,1400); r19 (4535925,168425,19296) versus (4898233,187497,19296); r199 (19264532,384488,10150) versus (22281024,466383,10150). The r7 multiplication loss is retained. Cache setup and decoder costs are separate. These are counted finite-panel quantities, with no timing or population inference. [pcc:unverified]

Primary P192. All24 prescribed27-factor responses match sealed Round63 on publicQ,mu_G,relation_lambda_G. The adapter now directly forms rational numerator polynomials with public beta/gamma, uses the cached denominator, and compares each of72 degree4 baby norms against direct pairs. It preserves the unique planted zero and all24 combined coefficient vectors. Selected query totals are56637 multiplications/1605 inversions/168 point-add calls versus60813/2037/168. The cache costs60 further multiplications for the3 targets. Both modes share54486 numerator-formation/norm multiplications and1296 inversions. Denominator work is1080/0 versus5256/432, including denominator-polynomial preparation in the legacy arm. Two output divisions per block cost144 multiplication calls and144 inversions. P192 counters exclude inverse internals and cannot be combined with the small-field counts. The selected27-factor windows are a prescribed subset, with no complete order-r P192 response executed. [pcc:unverified]

Validation. Scope and source were published before affected data, source358f52569fc31d582abba8c90caa0cd8595fec01. Candidate and legacy small and P192 science all exit0. A fresh source copy reproduces four science records and two count receipts byte-for-byte. Changed small calibration and P192 first-coefficient records reject101. No new build failure occurred. The unchanged Round63 legacy-library compilation timeout124 is preserved without retry and is not a current-head library-test pass. No cloud job, managed generator, role, quota or old freeze lane was changed. [pcc:unverified]

Next construction. Numerator formation and contraction dominate the P192 accounting. Investigate cancellation of a factor D in E0+s*E1 using the exact elliptic biquadratic addition identities. The successor must prove the polynomial identity and check reduced numerator norms against these sealed responses before assigning any reduced cost. [pcc:unverified]

Sources checked. Round63 paired-algebra-proof.md, Round64 resultant-proof.md and Round65 report.md; Stacks0BIE, https://stacks.math.columbia.edu/tag/0BIE (read2026-11-09), for multiplicative norms. KB tools and migrated wrapper unavailable. Standard ingredients are used; this component makes no novelty assertion. [pcc:unverified]